TILINGS OF W_M AND T_M BELOW THE THRESHOLD SCALE: A SIX-PIECE TEMPLATE

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Abstract. Fix coprime integers $2 \le e \le f-2$ and the triangle (the *tile*) with sides ef, f^2-e^2, f^2. The triangles W_M and T_M of scale M were known to be tiled by $M^2(2f^2-e^2)$ and $M^2(3f^2-e^2)$ tiles when $M \ge f$ (Paliy). We prove that W_M is tiled as soon as $Me \ge f$ and $(M-1)(f-e) \ge f$, equivalently $(f-e)\lfloor eM/f \rfloor \ge e+1$; for T_M the tile count is proved and the tiling follows from a gluing lemma imported from the earlier release. The proof is one layout of six pieces in a skew frame with integer coordinates; it has a written proof of about four pages and, independently, a Lean 4 proof. For $e < f/2$ and *assuming an imported exclusion theorem*, the condition is exact for W_M. Example: $W_4(3,5)$ is tiled by 656 tiles (open in the earlier release). A program count (computed, not proved) gives 26 138 new tile counts. The paper says which part is Lean, hand, computed or imported.

Key (everything needed to read §1.2). Coprime integers $2 \le e \le f-2$; $d = f-e$; tile sides $a = ef$, $b = f^2-e^2$, $c = f^2$; $\kappa = 2f^2-e^2$. A triangle of *scale* s has sides sa, sb, sc (tiled by s^2 tiles). W_M is the triangle OAB with $|OB| = Mfb$, $|OA| = Mf^3$, $|AB| = Me\kappa$; P is the point of AB with $OP = OB$. *Frame:* origin P, axes along PO and PB; a point is (ξ, y), a vector has squared length $\xi^2 + y^2 + \frac{e}{f}\xi y$; $O = (fbM, 0)$, $B = (0, ebM)$, $A = (0, -ef^2M)$ (all points of the layout have integer coordinates, Table 3). *Lattice lines:* with $\varphi = f\xi + ey$, the line ℓ_k is $\varphi = efbk$ ($k \in \mathbb{Z}$); it has direction $(e, -f)$ and meets PO at $\xi = ebk$ and AB at $y = fbk$. ℓ_t and ℓ_r are the two lattice lines on either side of B ($ft < eM < fr$), and the *strip* S is the part of OPB between them; its width is one tile-row ($= af$ along OB) when $f \nmid eM$. Lattice coordinates $u = \varphi/efb$, $w = (e\xi + fy)/efb$ tell on which side of a line a point is. $n = \lfloor eM/f \rfloor$; canonical $t = n - \delta$, $r = n+1$, $q = n$ ($\delta = 1$ iff $f \mid eM$). (H1) $Me \ge f$, (H2) $(M-1)d \ge f$.

1. Introduction

1.1. **The result.** A triangle Ω is *tiled by N copies of a triangle* tile if Ω is the union of N triangles with pairwise disjoint interiors, each congruent to tile (mirror images allowed, no edge-to-edge condition). An integer N is a *tile count* if some triangle is tiled by N copies of some triangle; Erdős asked which integers are tile counts (Problem 634 of `erdosproblems.com`). Paper W constructs tilings; nothing about primes is used or claimed (the tiles with $3\alpha + 2\beta = \pi$ are the ones on which the base-β approach of Beeson–Roşu and Paliy to the prime case rests; here each tiling of a target is only a new tile count).

For coprime integers $1 \le e < f$ put $a = ef$, $b = f^2 - e^2$, $c = f^2$; the triangle $\mathsf{tile}(e,f)$ with sides a, b, c has angles α, β, γ opposite a, b, c and $3\alpha + 2\beta = \pi$. Beeson–Roşu and Paliy define two target triangles, $W_M(e,f)$ and $T_M(e,f)$, for each integer scale $M \ge 1$ (Section 2.3); their area is $M^2(2f^2-e^2)$ resp. $M^2(3f^2-e^2)$ times the area of the tile. Before this paper both were known to be tiled for every $M \ge f$ (Paliy), and for $e \ge 2$ the expectation was that no smaller M works.

> **Theorem W** (Theorem 3.1). If $2 \le e \le f-2$, $\gcd(e,f) = 1$, $Me \ge f$ and $(M-1)(f-e) \ge f$, then $W_M(e,f)$ is tiled by $M^2(2f^2-e^2)$ copies of $\mathsf{tile}(e,f)$ and $T_M(e,f)$ by $M^2(3f^2-e^2)$.

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Example: $(e,f)=(2,5)$, $M=3$ gives a tiling of a triangle with sides $315,276,375$ by 414 copies of $(10,21,25)$; before, the smallest scale known was $M=f=5$. *What this does not do.* It does not settle which M fail when $e>f/2$ (Section 6), the members $e=1$ and $e=f-1$, or the prime case of Problem 634.

1.2. **Reader's guide.** *The proof in outline* (Section 4; every claim below is an equation or inequality a reader can check by hand; "scale s" means a triangle with sides sa, sb, sc, and a "tile-row" is a strip one scale-one tile wide). (1) Put the origin at the point P of the side AB with $OP=OB$, and use as axes the lines PO and PB (Section 2.4); in this skew frame the three vertices of W_M and all other points below have *integer* coordinates, and a vector (ξ,y) has squared length $\xi^2+y^2+\frac{e}{f}\xi y$. (2) Five directions occur; for each, one integer vector and its length are listed (Table 2), so every side length is read off as "integer $\times$ length of a direction vector". (3) The line OP cuts W_M into the triangle OAP, which is a tile of the integer scale fM, and a triangle OPB. (4) The lines ℓ_k ($k\in\mathbb{Z}$), parallel to a side of the tile, form a lattice; ℓ_t and ℓ_r are the two lattice lines on either side of B, and between them lies a strip S one tile-row wide. (5) Two cuts parallel to OB divide S into three pieces; with ℓ_t, ℓ_r and the line P_2P_3 they give six pieces, defined by inequalities. (6) Covering is a case distinction on the signs of y, $u-t$, $u-r$ and $w-q$; disjointness is: two pieces lie on opposite sides of one of six lines (Section 4.3). (7) Each piece is a triangle, a trapezoid or a parallelogram of integer scale; Table 5 lists its side vectors (Section 4.4). (8) Everything above needs only $ft<eM<fr$, $er<fq$, $eq<ft$ for the integer parameters t,r,q (Section 4.1); these four inequalities have a solution exactly when $(f-e)\lfloor eM/f\rfloor\ge e+1$, equivalently when $Me\ge f$ and $(M-1)(f-e)\ge f$ (Lemma 3.5, Proposition 4.1). (9) The tile counts of the six pieces add up to $M^2(2f^2-e^2)$ (one identity).

How the layout was found. A computer search (our engine, run on the $M<f$ cases left open in the earlier release) produced tilings whose pieces were always the same six shapes; the proof below says why: the bisector OP makes one big tile, and what is left near B is a strip that one row of lattice-line pieces fills.

What is proved where.

claim	source
$W_M(e,f)$ is tiled (Thm 3.1)	Lean `thmW` (`WTilable`); and the hand proof of Section 4, a different argument
$M^2(2f^2-e^2)$, $M^2(3f^2-e^2)$ are tile counts	Lean `thmW`
$T_M(e,f)$ is tiled	glue (I5): Lean, earlier release
(H1)$\wedge$(H2) $\Leftrightarrow$ $(f-e)\lfloor eM/f\rfloor\ge e+1$; the layout exists iff this holds	Lemma 3.5, Prop. 4.1: hand proofs (a few lines), exact enumeration; not Lean
Sharpness for $e<f/2$	"if": this paper; "only if": imported (I1)
Explicit tilings (414, 656, …)	exact checker ([COMPUTED, EXACT]), not Lean
26 138 new tile counts	computer count ([COMPUTED, EXACT]); each value is an instance of Theorem 3.1
$W_2(5,8)$, other open scales	see Section 6; evidence only

1.3. **Labels, credit, reading.** Every statement carries a label: [PROVED IN LEAN] a theorem of the accompanying Lean 4 library, checked by the kernel with the standard axioms only (`propext`, `Classical.choice`, `Quot.sound`), no `sorry`; the Lean statement was compared with the written one *inside this project only*; [PROVED BY HAND] a written proof in which every step is displayed; the displayed identities and inequalities of Sections 2–4 were re-checked by exact enumeration (Appendix A), no outside reader has checked them; [COMPUTED, EXACT] an exact finite computation (for an explicit tiling: exact coordinates and an exact checker); [CONJECTURE] unproved.

Credits are in Section 7; the work was done with an AI system (disclosed there). *Reading.* Ten minutes: this section, Section 2, the point table (Table 3), Figure 3 and Example 5.1. An hour: Sections 2–5. Appendix A ("verification aid") is not needed to follow the proof.

2. The frame and the notation

2.1. **Symbols (reference table).** Table 1 lists every symbol of this paper; each is also defined where it first appears.

Table 1. Symbols. This table is shared word for word with the companion paper [6].

symbol	meaning ([W] = Paper W; [C] = the companion paper)
used in both papers	
$e<f$	coprime positive integers defining the tile; $d=f-e$
a,b,c,κ	$a=ef$, $b=f^2-e^2$, $c=f^2$, $\kappa=b+c=2f^2-e^2$: the sides of the tile $\mathsf{tile}(e,f)$ and the sum $b+c$
α,β,γ	angles of the tile opposite a,b,c; $3\alpha+2\beta=\pi$, $\gamma=2\alpha+\beta$
scale s	a triangle with sides sa,sb,sc has *scale* s and is tiled by s^2 tiles (ladder subdivision, [W])
W_M, T_M	the two targets of [W]; M is the *scale* of the target; W_M is tiled by $M^2(2f^2-e^2)$ tiles, T_M by $M^2(3f^2-e^2)$
$M^*(e,f)$	threshold scale $\max(\lceil f/e\rceil,\lceil f/(f-e)\rceil+1)$ of Theorem W; (H1), (H2) are $Me\ge f$ and $(M-1)(f-e)\ge f$, together equivalent to $M\ge M^*(e,f)$
used in [W] only	
n,ρ,δ	$n=\lfloor eM/f\rfloor$, $\rho=eM-fn$, $\delta=1$ if $\rho=0$ and 0 otherwise
t,r,q	the three free integers of the six-piece layout; canonical choice $t=n-\delta$, $r=n+1$, $q=n$
(ξ,y), (u,w)	skew frame with origin P ($OP=OB$) and lattice coordinates $u=(f\xi+ey)/efb$, $w=(e\xi+fy)/efb$; $\ell_k=\{u=k\}$ (Section 2.3 of [W])
v,s_1,s_2,x,u_1,u_2	the integer scales of the pieces, $v=fM-er$, $s_1=f(r-t)$, $s_2=fr-eM$, $x=ft-eq$, $u_1=fq-et$, $u_2=fq-er$
(K)	the four inequalities $ft<eM<fr$, $er<fq$, $eq<ft$
$\pi_0,\dots,\pi_5$	the six pieces of the layout; $O,A,B,P,P_2,P_3,Q,R,B',C',V_2,X_4,U_1,U_2$ its points
used in [C] only	
B_g	the isosceles target with base angles α; g is its scale; $b\kappa g^2$ tiles
row (e,f)	the tile $\mathsf{tile}(e,f)$; $G(e,f)$ is the set of scales g for which $B_g(e,f)$ is tiled
$F_0,\dots,F_8$	the families of Conjecture C

2.2. **The tile and the ladder.** Throughout, $e<f$ are coprime positive integers and

$$a=ef,\qquad b=f^2-e^2,\qquad c=f^2,\qquad \kappa=b+c=2f^2-e^2,\qquad d=f-e.$$

The *tile* $\mathsf{tile}(e,f)$ has sides a,b,c (a triangle since $a+b-c=ed>0$); α,β,γ are its angles opposite a,b,c. One checks $c^2=a^2+bc$, and $3\alpha+2\beta=\pi$, $\gamma=2\alpha+\beta$. The only angle used below is

$$\cos\beta=\frac{a^2+c^2-b^2}{2ac}=\frac{e(3f^2-e^2)}{2f^3}.$$

Examples: $(e,f)=(2,5)$ gives the tile $(10,21,25)$; $(3,5)$ gives $(15,16,25)$; $(1,2)$ gives $(2,3,4)$.

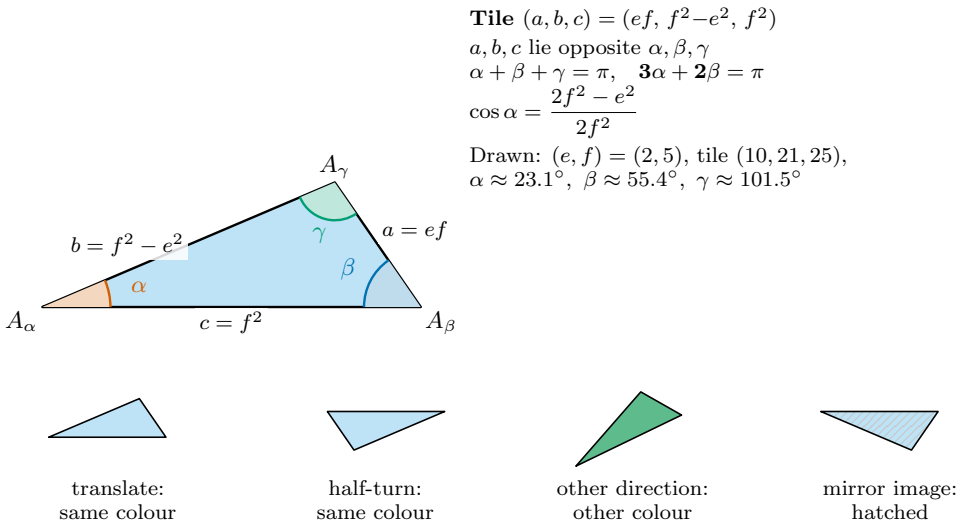


FIGURE 1. The tile tile(e,f) for $(e,f)=(2,5)$, the triangle $(10,21,25)$: sides $a=ef$, $b=f^2-e^2$, $c=f^2$ and angles $\alpha,\beta,\gamma=2\alpha+\beta$ opposite them; $3\alpha+2\beta=\pi$. In tiling figures, tiles of the same orientation (up to a half-turn) share a colour and mirror images are hatched.

A triangle with sides sa, sb, sc has *scale* s (a copy of the tile enlarged by s; by SSS it is determined by its side lengths).

Proposition 2.1 (Ladder subdivision; [PROVED IN LEAN]). *Let $s,u\geq 1$ be integers. A triangle of scale s is tiled by s^2 copies of* tile. *A parallelogram with sides sa and uc enclosing the angle β is tiled by $2su$ copies.*

Lean: `Subdivision.ladderDissection`, `Ladder.ladder` (earlier release, `thm:ladder`). *Why:* the lines parallel to the sides at integer steps cut a triangle of scale s into s^2 triangles of scale one, all congruent to the tile (half of them rotated by π); a parallelogram with sides a,c and angle β is two tiles, since its short diagonal has length b (law of cosines, $\cos\beta$ above), and a parallelogram with sides sa, uc is su of those. A *trapezoid* below is a triangle of scale s_1 minus a corner triangle of scale $s_2<s_1$ cut off by a line parallel to the opposite side; it is tiled by $s_1^2-s_2^2$ copies (the corner is the union of s_2^2 cells of the ladder).

2.3. **The targets.** For an integer $M\geq 1$ the two targets are the triangles

target	sides	angles (for orientation)	tiles by area
$W_M(e,f)$	$Mfb,\ Me\kappa,\ Mf^3$	2α (opposite $Me\kappa$), β (opposite Mfb), $\alpha+\beta$ (opposite Mf^3)	$M^2(2f^2-e^2)$
$T_M(e,f)$	$Mf^3,\ Mf^3,\ Me(3f^2-e^2)$	isosceles; apex 3α, base angles β	$M^2(3f^2-e^2)$

(The last column is the area of the target divided by the area of the tile, so any tiling has exactly that many tiles.) T_M is W_M with one tile of scale Mf glued on the side Mfb (Section 3.3); both are drawn in Figure 2. Name the vertices of W_M so that $|OB|=Mfb$, $|OA|=Mf^3$, $|AB|=Me\kappa$; the angle at O is 2α.

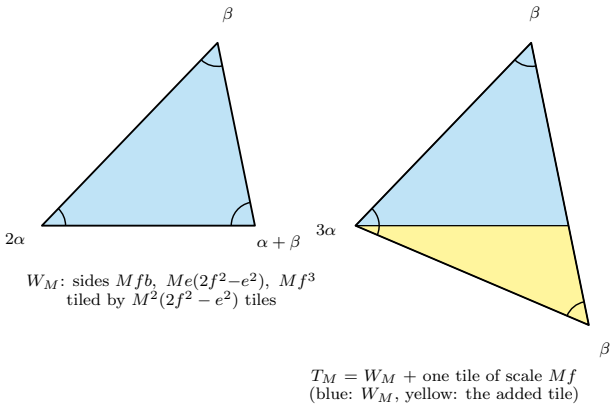


FIGURE 2. The two targets, with the proportions of $(e, f) = (2, 5)$, $M = 1$. Left: W_M with angles 2α (at O), β (at A), $\alpha+\beta$ (at B). Right: T_M, obtained from W_M by adding one tile of scale Mf (yellow) along the side Mfb.

2.4. **The frame.** Let P be the point of AB with $|OP| = |OB|$. Then $|PA| = Mef^2$ and $|PB| = |AB| - |PA| = Me\kappa - Mef^2 = Meb$ (so P lies between A and B and $PB = ebM$, $PA = ef^2M$). In the isosceles triangle OPB the angle at P equals the angle at B, and $\cos\angle OPB = \frac{|PB|/2}{|OB|} = \frac{e}{2f}$. *The frame.* Use P as origin, the unit vector from P towards O as first axis and the unit vector from P towards B as second axis. A point has coordinates (ξ, y), and a vector (ξ, y) has (law of cosines)

$$\text{length}^2 = \xi^2 + y^2 + \tfrac{e}{f}\,\xi y. \tag{1}$$

Then $O = (fbM, 0)$, $B = (0, ebM)$, $A = (0, -ef^2M)$, and (1) gives $|OB|^2 = b^2M^2(f^2 + e^2 - e^2) = (fbM)^2$, $|OA|^2 = M^2f^2(b^2 + e^2f^2 + e^2b) = (f^3M)^2$ (because $b^2 + e^2b + e^2f^2 = f^4$), $|AB| = ef^2M + ebM = Me\kappa$: this is W_M. *Five directions.* Every side below is parallel to one of five lines, with primitive integer vector and length (by (1)): *Lattice coordinates.* Put $\varphi = f\xi + ey$

TABLE 2. The five directions of the layout. The length of the vector is exact; a segment in that direction with integer coordinates is an integer multiple of it.

line	vector	length	check
OP, first axis	$(1, 0)$	1	
AB, second axis	$(0, 1)$	1	
OB and the cuts QV_2, U_1U_2	$(f, -e)$	f	$f^2 + e^2 - e^2 = f^2$
the lines ℓ_k and QU_1, V_2U_2, P_2U_1	$(e, -f)$	f	$e^2 + f^2 - e^2 = f^2$
OA and P_2P_3	(b, ef)	f^2	$b^2 + e^2f^2 + e^2b = f^4$

and $\psi = e\xi + fy$, and

$$u = \frac{\varphi}{efb}, \qquad w = \frac{\psi}{efb}, \qquad \text{so that} \qquad \xi = ef(fu - ew), \quad y = ef(fw - eu).$$

For an integer k the line ℓ_k is $\{u = k\}$; it has direction $(e, -f)$. It meets PO ($y = 0$) at $\xi = ebk$ and AB ($\xi = 0$) at $y = fbk$. The side OB is $\{w = M\}$, the line AB is $\{\xi = 0\} = \{ew = fu\}$, the line OP is $\{y = 0\} = \{eu = fw\}$. u, w are used only to say which side of a line a point lies on; lengths are always computed in (ξ, y).

2.5. **The points of the layout.** The layout depends on the scale M and on three integers t, r, q (chosen in Section 4.1); the integer *scales* of its pieces are

$$v = fM - er, \quad s_1 = f(r-t), \quad s_2 = fr - eM, \quad x = ft - eq, \quad u_1 = fq - et, \quad u_2 = fq - er. \tag{2}$$

(Every entry of the (u, w) columns follows from $u = \varphi/efb$, $w = \psi/efb$; for instance $u(U_1) = \big(f \cdot ef(ft-eq) + e \cdot ef(fq-et)\big)/efb = \big(ef \cdot bt\big)/efb = t$.) The earlier version of this paper described the same points by fourteen complex numbers; the dictionary is in Appendix A.3.

TABLE 3. The points, in the frame (ξ, y) (integers) and in lattice coordinates (u, w). R and X_4 are apexes of the corners that are removed from two pieces, not vertices of pieces.

point	(ξ, y)	(u, w)	point	(ξ, y)	(u, w)
P	$(0, 0)$	$(0, 0)$	V_2	$\big(ef^2(r-t),\ f(f^2t - e^2r)\big)$	$(r,\ ft/e)$
O	$(fbM, 0)$	$(fM/e,\ M)$	X_4	$(fbq, 0)$	$(fq/e,\ q)$
B	$(0, ebM)$	$(eM/f,\ M)$	U_1	$\big(ef(ft-eq),\ ef(fq-et)\big)$	$(t,\ q)$
A	$(0, -ef^2M)$	$(-\frac{efM}{b}, -\frac{f^2M}{b})$	U_2	$\big(ef(fr-eq),\ ef(fq-er)\big)$	$(r,\ q)$
P_2	$(bet, 0)$	$(t,\ et/f)$	C'	$(ber, 0)$	$(r,\ er/f)$
P_3	$(0, -e^2ft)$	$(-\frac{e^2t}{b}, -\frac{eft}{b})$	B'	$\big(ef(fr-eM),\ ef(fM-er)\big)$	$(r,\ M)$
Q	$(0, fbt)$	$(t,\ ft/e)$	R	$(0, fbr)$	$(r,\ fr/e)$

3. Theorem W and its consequences

3.1. **Statement.**

Theorem 3.1 (Theorem W; W half [PROVED IN LEAN] and [PROVED BY HAND]; count of T_M [PROVED IN LEAN]; tiling of T_M via the imported glue)**.** *Let e, f be coprime integers with $2 \le e \le f-2$ and let $M \ge 1$ be an integer with*

$$\textit{(H1)} \quad Me \ge f, \qquad\qquad \textit{(H2)} \quad (M-1)(f-e) \ge f.$$

Then $W_M(e, f)$ is tiled by $M^2(2f^2 - e^2)$ copies of tile(e, f)*, and $T_M(e, f)$ is tiled by $M^2(3f^2 - e^2)$ copies.*

Put $M^*(e, f) = \max\big(\lceil f/e \rceil,\ \lceil f/(f-e) \rceil + 1\big)$; then (H1) and (H2) together say $M \ge M^*(e, f)$.

Exactly what is Lean. Lean module `Draft/ThmW.lean` (with 19 helper modules, aggregate `Erdos634.AllThmW`), theorem `Draft.ThmW.thmW`: for natural numbers e, f, M with `Nat.Coprime e f`, $2 \le e$, $e + 2 \le f$, $f \le M \cdot e$ and $f \le (M-1) \cdot (f-e)$, the conclusion is

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WTilable e f M ∧ TileCount ((2*f^2-e^2)*M^2) ∧ TileCount (M^2*(3*f^2-e^2)).
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`WTilable e f M` says: there is a dissection into congruent triangles, each congruent to the triangle with sides $ef,\ f^2 - e^2,\ f^2$, of a triangle with sides $Mf(f^2 - e^2)$, $Me(2f^2 - e^2)$, Mf^3. So Lean states the *tiling* for W_M; for T_M it states only that the number is a tile count, and the tiling of T_M itself is the glue (I5), a Lean theorem of the earlier release (Section 3.3). Axioms: `propext`, `Classical.choice`, `Quot.sound`. The Lean statement was compared with the one above inside the project (verdict: match). Hypotheses are witnessed by instances `inst_414`, `inst_656`, `inst_801` (and others), so the Lean theorem is not vacuous. *The Lean proof does not follow Section 4*: it verifies, for the layout in complex coordinates, a table of about 100 cross products factored into positive factors, with 143 generated sign lemmas (Appendix A.3). Section 4 is a second, independent proof of the W half, written for a reader; it is not formalised.

First instances (the last column is the smallest scale known before, namely f):

(e, f)	M^*	first tiled W_{M^*}	number of tiles	smallest M known before
$(2, 5)$	3	$W_3(2, 5)$	414	5
$(3, 5)$	4	$W_4(3, 5)$	656	5
$(3, 7)$	3	$W_3(3, 7)$	801	7
$(2, 11)$	6	$W_6(2, 11)$	8568	11
$(9, 11)$	7	$W_7(9, 11)$	7889	11

Thirteen targets with $M < f$ were found by a bounded computer search before the proof was known (Appendix B); they are only the output of that one search. The theorem has many more instances with $M < f$, for example $(e, f, M) = (4, 9, 3)$ and $(3, 8, 3)$.

3.2. **Consequences.**

Corollary 3.2 (Sharpness for $e < f/2$, assuming the import (I1); "if": Theorem 3.1 [PROVED IN LEAN] plus the arithmetic below; "only if": imported (I1))**.** *Let* $\gcd(e, f) = 1$ *and* $2 \le e < f/2$. *Then* $W_M(e, f)$ *is tiled if and only if* $Me \ge f$.

Proof. "Only if" is (I1) (Section 3.4). "If": $e < f/2$ gives $f - e > f/2$, so $f/(f - e) < 2$ and $\lceil f/(f - e) \rceil + 1 = 3$, while $f/e > 2$ gives $\lceil f/e \rceil \ge 3$; so $M^* = \lceil f/e \rceil$. Moreover (H1) implies (H2): (H1) gives $M \ge 3$, so $(M - 1)(f - e) \ge 2(f - e) > f$. Apply Theorem 3.1. □

For $e < f/2$ the threshold is therefore *exactly* $\lceil f/e \rceil$, smaller than f as soon as $e \ge 2$. (For $e = 1$ it is f, by (I3).) The "only if" half is not proved in this paper; it is only as strong as the import (I1).

Corollary 3.3 (An open case; [PROVED IN LEAN] for the tiling, [COMPUTED, EXACT] for the data)**.** $W_4(3, 5)$*, a triangle with sides* 320, 492, 500*, is tiled by* 656 *copies of* $(15, 16, 25)$.

This is Theorem 3.1 for $(e, f, M) = (3, 5, 4)$ ($M^*(3, 5) = \max(2, 4) = 4$). A list of the 656 tiles in exact coordinates is shipped (Appendix B) and verified by an exact checker.

Remark 3.4 (An earlier expectation that is false)**.** (a) *The threshold conjecture "B1".* Our earlier notes expected that, for $e \ge 2$, neither W_M nor T_M is tiled for $M < f$; the earlier paper [4] recorded as an observation that every case decided so far has threshold f, and listed $W_4(3, 5)$ among the open cases. Corollary 3.3 is a counterexample ($4 < 5$), and Theorem 3.1 gives infinitely many. The first statement was an expectation of our working notes and an observation in the earlier paper, not a theorem; no earlier theorem is affected.

3.3. **The T half.** Glue to the side Mfb of W_M (the side OB) one copy of the tile at scale Mf, with its side Mfb on that side and its third vertex Z on the line AB beyond B, so that $BZ = Mef^2$. The result is the isosceles triangle with legs Mf^3 and base $Me(3f^2 - e^2)$, that is T_M, and its tile count is $M^2(2f^2 - e^2) + (Mf)^2 = M^2(3f^2 - e^2)$. This is (I5). The Lean theorem `Draft.ThmW.thmW` contains the *count* for T_M, obtained from the tiling of W_M by `DebtB.counts_of_wTilable`, which needs besides the tiling only $M(f - e) \ge f$; (H2) implies it. The converse (a tiling of T_M gives one of W_M) is not known, and the proved exclusions for T_M are weaker ((I1): only $2Me < f$). Hence the label of the T half: tiling of W_M [PROVED IN LEAN] [PROVED BY HAND]; tile count of T_M [PROVED IN LEAN]; tiling of T_M by the imported glue [PROVED IN LEAN] in the earlier release.

3.4. **Five facts imported from the earlier release.** Five facts about the tile and the targets are Lean theorems of the earlier release [4, 5] and are used as stated, not reproved. *Tiled* means tiled by copies of $\mathsf{tile}(e, f)$ with the same (e, f).

(I1) (`cor:ebelow`(a), (b), and the case $e = 1$) If $Me < f$ then $W_M(e,f)$ is not tiled; if $2Me < f$ then $T_M(e,f)$ is not tiled; for $e = 1$, $T_M(1,f)$ is not tiled if $2M \le f$. [All coprime $1 \le e < f$.]
(I2) (`cor:thickw`(a)) If $M(f-e) \le e$ then $W_M(e,f)$ is not tiled.
(I3) (`cor:below`, `cor:thickw`(b)) For $e = 1$ and for $e = f-1$: $W_M(e,f)$ is tiled if and only if $M \ge f$.
(I4) (Paliy; formalised there) For every $M \ge f$, $W_M(e,f)$ and $T_M(e,f)$ are tiled.
(I5) (the glue; Lean `CollarG.betaTilable_of_wTilable` and `DebtB.counts_of_wTilable`) If $W_M(e,f)$ is tiled then $T_M(e,f)$ is tiled; and if $W_{M_0}(e,f)$ is tiled and $M_0(f-e) \ge f$, then for every $M \ge M_0$ both W_M and T_M are tiled and $M^2(2f^2-e^2)$, $M^2(3f^2-e^2)$ are tile counts.

They are used only outside the proof of the W half: (I1) for Corollary 3.2 and to say what is open; (I2), (I3) to say what is open and in Remark 3.4; (I4) to place the theorem against what was known; (I5) for the T half. The proof of the W half (Section 4) uses none of them.

3.5. **An equivalent form of the hypotheses.** Write $n = \lfloor eM/f \rfloor$, $\rho = eM - fn \in [0,f)$ and $\delta = 1$ if $\rho = 0$, $\delta = 0$ otherwise.

Lemma 3.5 ([PROVED BY HAND]; not in Lean)**.** *Let $2 \le e \le f-2$, $\gcd(e,f) = 1$, $d = f-e$, $M \ge 1$. Then (H1) and (H2) hold if and only if $d\,n \ge e+1$.*

Proof. First, $ed \ge f+1$: with $f = e+d$ we have $ed - f = (e-1)(d-1) - 1$, and $(e-1)(d-1) \ge 2$ because $e, d \ge 2$ and $e = d = 2$ would give $f = 4$, not coprime to $e = 2$. $(*)$

($\Leftarrow$) Suppose $dn \ge e+1$. Then $n \ge 1$, i.e. $eM \ge f$: (H1). Suppose (H2) failed: $(M-1)d \le f-1$. Then $dM \le f+d-1 < 2f$, so $s := \lceil dM/f \rceil \in \{1,2\}$, and $n = \lfloor M - dM/f \rfloor = M - s$. If $s = 1$ then $dM \le f$ and $dn = d(M-1) \le f-d = e$; if $s = 2$ then $dn = d(M-2) \le f+d-1-2d = e-1$. Either way $dn \le e$, a contradiction.

($\Rightarrow$) Suppose (H1), (H2). If $d \ge e+1$: $n \ge 1$ by (H1), so $dn \ge d \ge e+1$. If $d \le e$: $d \nmid f$ (else $d \mid e$, so $d = 1$), so $f = dh + j$ with $0 < j < d$, and $h \ge 2$ because $f = e + d \ge 2d$; then $e = f - d = d(h-1) + j$. (H2) gives $M - 1 \ge \lceil f/d \rceil = h+1$, so $M \ge h+2$, and

$$e(h+2) - hf = e(h+2) - h(e+d) = 2e - hd = d(h-2) + 2j > 0,$$

so $eM \ge e(h+2) > hf$, hence $n \ge h$ and $dn \ge dh = f - j \ge f - d + 1 = e+1$. □

Corollary 3.6 (Theorem W, restated; [PROVED BY HAND] (Lemma 3.5) on top of [PROVED IN LEAN])**.** *For coprime $2 \le e \le f-2$ and $M \ge 1$ with $(f-e)\lfloor eM/f \rfloor \ge e+1$, the target $W_M(e,f)$ is tiled by $M^2(2f^2-e^2)$ tiles and $T_M(e,f)$ by $M^2(3f^2-e^2)$ tiles. The tile counts of the six pieces of Section 4 are, for $\delta = 0$,*

$$f^2M^2 - e^2n^2, \quad (fM - en - e)^2, \quad f^2n^2, \quad \rho(2f - \rho), \quad 2f\,dn, \quad e(2dn - e).$$

The hypotheses of the Lean theorem stay (H1), (H2); the lemma is proved here, by hand only. Consequently $M^*(e,f)$ is the least M with $d\lfloor eM/f \rfloor \ge e+1$.

4. Proof of the W half of Theorem W

Standing assumptions: e, f coprime, $2 \le e \le f-2$, $d = f - e \ge 2$, $M \ge 1$; t, r, q integers.

4.1. **The four inequalities.** We use the following condition on (t,r,q):

$$\text{(K)} \qquad ft < eM < fr, \qquad er < fq, \qquad eq < ft. \tag{3}$$

In lattice coordinates, (K) says: on the line AB the point B lies between the lattice lines ℓ_t and ℓ_r, and on the lines ℓ_t, ℓ_r the level q lies in the window $er/f < q < ft/e$. Consequences used below (each a one-line derivation):

$$t < r, \quad t \geq 1, \quad q \geq 1, \quad q < M,$$
$$e^2 r < f^2 t, \quad er < fM, \quad et < fq, \quad et < fM. \tag{4}$$

(Derivations: $ft < eM < fr$ gives $t < r$; $er > 0$ and $er < fq$ give $q \geq 1$; $eq < ft < eM$ gives $q < M$; $eq > 0$ gives $t \geq 1$; $e^2 r < efq < f^2 t$; $er < fq < fM$; $et < er < fq$; $et < eM < fM$.) All six scales (2) are positive and $s_1 > s_2$, $u_1 > u_2$: $v = fM - er > 0$; $s_1 = f(r-t) > 0$; $s_2 = fr - eM > 0$; $s_1 - s_2 = eM - ft > 0$; $x = ft - eq > 0$; $u_2 = fq - er > 0$; $u_1 - u_2 = e(r-t) > 0$.

Proposition 4.1 (the layout exists exactly when $dn \geq e+1$; [PROVED BY HAND], [COMPUTED, EXACT] for the enumeration)**.** *For coprime $2 \leq e \leq f-2$ and $M \geq 1$ the following are equivalent: (i) there are integers t, r, q with (K); (ii) $dn \geq e+1$ where $n = \lfloor eM/f \rfloor$; (iii) (H1) and (H2). If they hold, the triple $t = n - \delta$, $r = n+1$, $q = n$ satisfies (K). Every triple satisfying (K) has $t \leq n - \delta$ and $r \geq n+1$.*

Proof. (ii)⇔(iii) is Lemma 3.5. (ii)⇒(i) for the stated triple: $n \geq 1$ by (ii). $ft < eM$: if $\delta = 0$ then $\rho > 0$ and $fn < fn + \rho = eM$; if $\delta = 1$ then $f(n-1) < fn = eM$. $eM < fr$: $eM = fn + \rho < fn + f$. $er < fq$: $e(n+1) < fn \Leftrightarrow dn > e$, which is (ii). $eq < ft$: if $\delta = 0$, $en < fn$; if $\delta = 1$ then $f \mid eM$, so $f \mid M$ (coprimality), $M = fm'$, $n = em'$ and $dn \geq de \geq f+1$ by $(*)$, which is $en < f(n-1)$. (i)⇒(ii) and the last sentence: from $ft < eM$ we get $t < eM/f < n+1$, so $t \leq n$, and $t \leq n-1$ if $\rho = 0$; from $eM < fr$ we get $r > n$, so $r \geq n+1$. Suppose $dn \leq e$, that is $fn \leq e(n+1)$. Then $fq > er \geq e(n+1) \geq fn$, so $q \geq n+1$, and $eq \geq e(n+1) \geq fn \geq ft > eq$, a contradiction. □

So the layout below exists for *no* integers t, r, q when $dn \leq e$; this is a statement about this layout, not about tilings (Section 6). [COMPUTED, EXACT]: for $f \leq 12$ and all $M < 3f$, all integer triples with $0 \leq t < r \leq fM/e + 2$, $0 \leq q \leq M+2$ were enumerated; (K) (and the sixteen named factors of the older proof, Appendix A.3) hold for the same triples; no discrepancy.

Remark 4.2 (why these values of t, r, q)**.** Only $t \leq n - \delta$ and $r \geq n+1$ are forced (by $ft < eM < fr$). The admissible triples are not unique: for $W_3(2,5)$ the triples $(t,r,q) = (1,2,1), (1,2,2), (1,3,2), (1,4,2)$ all satisfy (K). The canonical triple $(n - \delta, n+1, n)$ is the extremal admissible one: it has the narrowest strip, and it is admissible whenever any triple is (Proposition 4.1). Nothing else is claimed.

4.2. **The six pieces.** Fix (t,r,q) with (K). The six pieces are the following regions, written with the lattice coordinates (u,w) and (ξ, y) of Section 2.4; $O, A, B, \ldots$ are the points of Table 3. The line $\{ef\xi - by = \text{const}\}$ is parallel to OA (direction (b, ef)); the one with constant $e^2 fbt$ passes through P_2 and P_3. Figure 3 shows the six pieces in the true shape of the frame for $W_3(2,5)$, with the integer lengths of the edges. Three of the cutting lines are ℓ_t, ℓ_r and the horizontal (w-constant, direction $(f, -e)$) lines $w = ft/e$ (through Q) and $w = q$ (through U_1, X_4).

4.3. **Step 1: the pieces cover W_M and overlap only in boundaries.** Write $\Omega = OAB = \{\xi \geq 0,\ w \leq M,\ ef\xi - by \leq ef^2 bM\}$ (the three sides AB, OB, OA).

Vertices. Each piece is the polygon whose vertices are listed in Table 4; that the listed points are the corners of the half-plane region is read off the lattice coordinates of Table 3. For π_4: the region is the rectangle $[t,r] \times [q, ft/e]$ in the (u,w)-plane, and $U_1 = (t,q)$, $U_2 = (r,q)$, $V_2 = (r, ft/e)$, $Q = (t, ft/e)$; it is non-degenerate since $q < ft/e$ by (K). For π_5: $P_2 = (t, et/f)$, $U_1 = (t,q)$, $U_2 = (r,q)$, $C' = (r, er/f)$, a trapezoid with two sides on $u = t$, $u = r$, since $et/f < er/f < q$ by (4). For π_3: it lies between the lines $w = ft/e$ and $w = M$ (note

TABLE 4. The six pieces as intersections of half-planes; vertices; and the shape (Section 4.4).

piece	region	vertices	shape and tile count
π_0	$y \le 0$, $\xi \ge 0$, $e^2fbt \le ef\xi - by \le ef^2bM$	A, O, P_2, P_3	trapezoid: OAP (scale fM) minus corner PP_2P_3 (scale et); $f^2M^2 - e^2t^2$
π_1	$y \ge 0$, $u \ge r$, $w \le M$	O, B', C'	triangle, scale v; v^2
π_2	$\xi \ge 0$, $u \le t$, $ef\xi - by \le e^2fbt$	P_3, P_2, Q	triangle, scale ft; f^2t^2
π_3	$\xi \ge 0$, $u \le r$, $ft/e \le w \le M$	B, Q, V_2, B'	trapezoid: RQV_2 (scale s_1) minus corner RBB' (scale s_2); $s_1^2 - s_2^2$
π_4	$t \le u \le r$, $q \le w \le ft/e$	Q, U_1, U_2, V_2	parallelogram $as_1 \times cx$, angle β; $2s_1x$
π_5	$y \ge 0$, $t \le u \le r$, $w \le q$	U_1, P_2, C', U_2	trapezoid: $X_4U_1P_2$ (scale u_1) minus corner $X_4C'U_2$ (scale u_2); $u_1^2 - u_2^2$

$ft/e < M$) and between the line AB (through $Q = (t, ft/e)$, $B = (eM/f, M)$) and the line $u = r$ ($V_2 = (r, ft/e)$, $B' = (r, M)$), and $eM/f < r$. For π_1: the triangle with vertices $O = (fM/e, M)$, $B' = (r, M)$, $C' = (r, er/f)$, non-degenerate since $eM/f < r < fM/e$ (the right inequality is $er < fM$). For π_2 and π_0: the corner P, P_2, P_3 and the lines $P_2P_3 \parallel OA$, AB, ℓ_t.

Containment. Ω is convex, so it suffices that every vertex lies on a segment joining points of Ω:

$$P_2, C' \in [P, O], \quad P_3 \in [A, P], \quad Q \in [P, B], \quad B' \in [B, O], \quad V_2, U_2 \in [C', B'], \quad U_1 \in [P_2, Q].$$

In coordinates these are, in this order: $0 \le t, r$ and $bet, ber \le fbM$ (as $et, er < fM$); $0 \le e^2ft \le ef^2M$ (as $et < fM$); $0 \le fbt \le ebM$ (as $ft < eM$); $eM/f \le r \le fM/e$ on the line $w = M$ (as $eM < fr$, $er < fM$); on the line $u = r$ the levels $w(V_2) = ft/e$ and $w(U_2) = q$ lie in $[er/f, M]$ (as $e^2r < f^2t$, $ft < eM$, $er < fq$, $q < M$); on $u = t$ the level $w(U_1) = q$ lies in $[et/f, ft/e]$ (as $et < fq$, $eq < ft$). All are (3), (4).

Covering. Let $X \in \Omega$.

- If $y \le 0$: X lies in π_0 if $ef\xi - by \ge e^2fbt$. Otherwise $ef\xi \le e^2fbt + by \le e^2fbt$, so $\xi \le ebt$ and $f\xi + ey \le efbt$, i.e. $u \le t$, and $X \in \pi_2$.
- If $y \ge 0$ and $u \le t$: $f\xi = efb\,u - ey \le efbt$, so $ef\xi - by \le e^2fbt$, and $X \in \pi_2$.
- If $u \ge r$ (and $y \ge 0$, $w \le M$): $X \in \pi_1$.
- If $y \ge 0$ and $t < u < r$: according as $w \ge ft/e$, $q \le w \le ft/e$, $w \le q$, the point lies in π_3, π_4, π_5.

So $\Omega \subseteq \bigcup \pi_i$, and by containment $\Omega = \bigcup \pi_i$.

Disjointness, pair by pair. For each pair the separating line, and the reason each piece lies on its side:

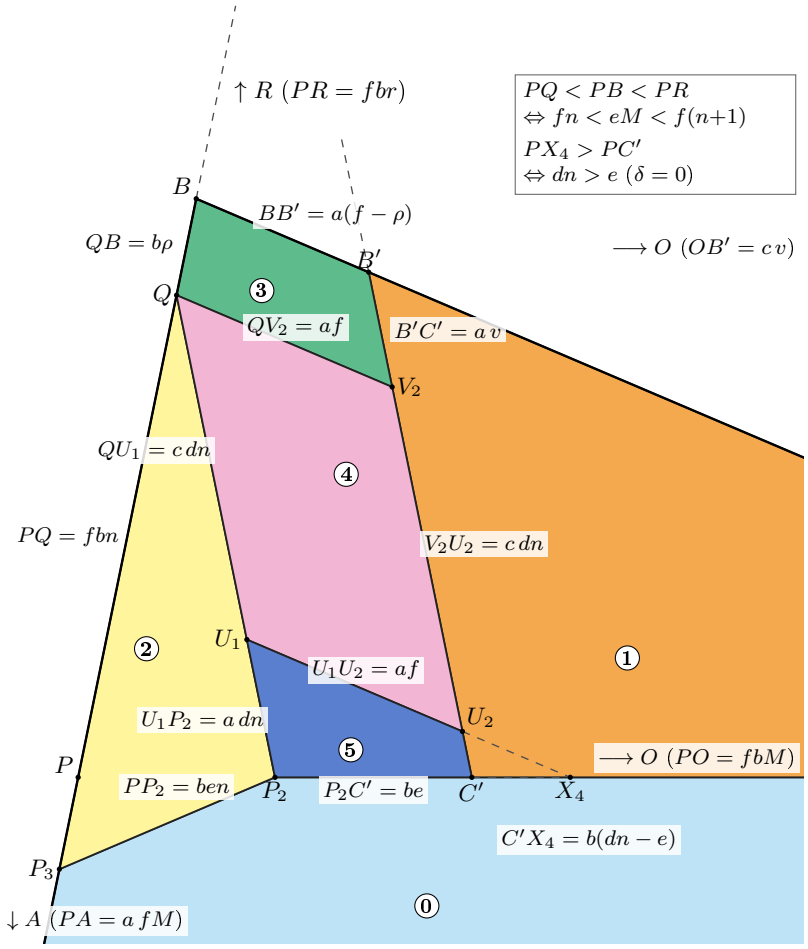


FIGURE 3. The six pieces of $W_3(2,5)$ in the true shape of the skew frame (angle at P is $\arccos\frac{1}{5}$), enlarged near P; $\delta = 0$, $(t,r,q) = (1,2,1)$, $n = 1$, $\rho = 1$. Pieces are numbered 0–5 as in Table 4. Edge labels are exact lengths: $a = 10$, $b = 21$, $c = 25$, $d = 3$, $f = 5$, $e = 2$, $M = 3$. The inequalities in the box are the two things the layout needs: the point B lies between Q and R (this fixes t and r), and the apex X_4 lies at or beyond C' on PO (this makes π_5 non-empty; $PX_4 - PC' = b(dn - e)$). O, A, R lie outside the drawing; $PO = fbM$, $PA = a\,fM$.

pairs	line	reason
$\pi_0 \mid \pi_1, \pi_3, \pi_4, \pi_5$	$y = 0$	π_0: $y \le 0$ by definition; π_1, π_5: $y \ge 0$ by definition; π_3, π_4: $w \ge eu/f$ since $w \ge q \ge er/f \ge eu/f$ (or $w \ge ft/e \ge er/f$, using $e^2 r < f^2 t$)
$\pi_0 \mid \pi_2$	$ef\xi - by = e^2 fbt$	the definitions
$\pi_2 \mid \pi_1, \pi_3, \pi_4, \pi_5$	$u = t$	π_2: $u \le t$; π_1 $(u \ge r)$, π_4, π_5 $(u \ge t)$; π_3: $u \ge ew/f \ge t$
$\pi_1 \mid \pi_3, \pi_4, \pi_5$	$u = r$	π_1: $u \ge r$; the others $u \le r$
$\pi_3 \mid \pi_4, \pi_5$	$w = ft/e$	π_3: $w \ge ft/e$; π_4, π_5: $w \le ft/e$ (as $q \le ft/e$)
$\pi_4 \mid \pi_5$	$w = q$	π_4: $w \ge q$; π_5: $w \le q$

4.4. **Step 2: each piece is a union of tiles.** Table 5 lists the side vectors of every triangle used, as an integer times a primitive vector of Table 2; each line is a subtraction of two rows of Table 3. By (1) the length of a side is that integer times the length of the primitive vector, so the triangle has the stated scale (SSS). *Consequences.* (a) OAP is a tile of scale fM and PP_2P_3

TABLE 5. Side vectors. In a triangle of scale s the three sides have lengths sa, sb, sc (listed in the order of the vectors).

triangle (scale)	side vectors	lengths
OAP (fM)	$O-P=fbM\,(1,0)$; $P-A=ef^2M\,(0,1)$; $A-O=-fM\,(b,ef)$	bfM; afM; cfM
PP_2P_3 (et)	$P_2-P=bet\,(1,0)$; $P-P_3=e^2ft\,(0,1)$; $P_3-P_2=-et\,(b,ef)$	$b{\cdot}et$; $a{\cdot}et$; $c{\cdot}et$
P_3P_2Q (ft)	$Q-P_3=f^3t\,(0,1)$; $Q-P_2=-bt\,(e,-f)$; $P_2-P_3=et\,(b,ef)$	$c{\cdot}ft$; $b{\cdot}ft$; $a{\cdot}ft$
$OB'C'$ (v)	$C'-O=-bv\,(1,0)$; $B'-O=-fv\,(f,-e)$; $B'-C'=-ev\,(e,-f)$	$b{\cdot}v$; $c{\cdot}v$; $a{\cdot}v$
RQV_2 (s_1)	$R-Q=fb(r-t)\,(0,1)$; $V_2-Q=es_1\,(f,-e)$; $V_2-R=fs_1\,(e,-f)$	$b{\cdot}s_1$; $a{\cdot}s_1$; $c{\cdot}s_1$
RBB' (s_2)	$R-B=bs_2\,(0,1)$; $B'-R=fs_2\,(e,-f)$; $B'-B\parallel OB$	$b{\cdot}s_2$; $c{\cdot}s_2$
$X_4U_1P_2$ (u_1)	$P_2-X_4=-bu_1\,(1,0)$; $U_1-X_4=-fu_1\,(f,-e)$; $P_2-U_1=eu_1\,(e,-f)$	$b{\cdot}u_1$; $c{\cdot}u_1$; $a{\cdot}u_1$
$X_4C'U_2$ (u_2)	$C'-X_4=-bu_2\,(1,0)$; $U_2-X_4=-fu_2\,(f,-e)$; $U_2-C'=-eu_2\,(e,-f)$	$b{\cdot}u_2$; $c{\cdot}u_2$; $a{\cdot}u_2$

is a corner of it cut off by the line $P_2P_3 \parallel OA$ at the integer step et: π_0 is a union of $f^2M^2-e^2t^2$ tiles (Prop. 2.1). (b) π_1 and π_2 are triangles of scales v and ft. (c) RQV_2 has scale s_1 and RBB' is its corner at R of scale s_2 (B on RQ, B' on RV_2, $BB' \parallel QV_2$, both $\parallel OB$): π_3 is a union of $s_1^2-s_2^2$ tiles. (d) $X_4U_1P_2$ has scale u_1 and $X_4C'U_2$ is its corner of scale u_2 ($C'U_2 \parallel P_2U_1$): π_5 is a union of $u_1^2-u_2^2$ tiles. (e) π_4 has the sides $U_1-Q=fx\,(e,-f)$ and $V_2-Q=es_1\,(f,-e)$, of lengths cx and as_1, and $U_2-V_2=U_1-Q$, $U_2-U_1=V_2-Q$, so it is a parallelogram; the angle at Q between $(f,-e)$ and $(e,-f)$ has cosine

$$\frac{\langle (f,-e),(e,-f)\rangle}{f\cdot f}=\frac{2ef+\frac{e}{2f}\left(-f^2-e^2\right)}{f^2}=\frac{e(3f^2-e^2)}{2f^3}=\cos\beta,$$

where $\langle(\xi,y),(\xi',y')\rangle=\xi\xi'+yy'+\frac{e}{2f}(\xi y'+y\xi')$ is the bilinear form of (1); so the angle is β and, by Proposition 2.1, π_4 is a union of $2s_1x$ tiles.

4.5. **Step 3: the count, and the end of the proof.** By Steps 1 and 2, W_M is the union of six pieces with disjoint interiors, each a union of tiles with the counts of Table 4; so W_M is tiled by their sum. That sum is $M^2\kappa$: the identity

$$(f^2M^2-e^2t^2)+v^2+f^2t^2+(s_1^2-s_2^2)+2s_1x+(u_1^2-u_2^2)=M^2(2f^2-e^2)$$

is a polynomial identity in e,f,M,t,r,q (expand with (2); checked by computer algebra, Appendix A). It is also forced by area: the tiles are disjoint, cover W_M, and all have the area of the tile. For $\delta=0$ and the canonical triple it reads, with $eM=fn+\rho$, $f^2M^2-e^2n^2+(fM-en-e)^2+f^2n^2+\rho(2f-\rho)+2f\,dn+e(2dn-e)=M^2\kappa$. The canonical triple of Proposition 4.1 satisfies (K) under (H1), (H2). This proves the W half of Theorem 3.1. The T half is Section 3.3. □

Remark (where the hypotheses enter). (H1) gives $n \geq 1$; (H2) enters only through $dn \geq e+1$, that is the single inequality $er < fq$ with $q = n$, $r = n+1$: the apex X_4 of the corner removed from π_5 lies on PO at or beyond C'. Everything else in (K) holds for all $M \geq 1$ with $dn \geq 1$.

Remark. A shorter alternative to Step 1 (a winding-number chain lemma, audited inside the project, not in Lean) is given in Appendix A.2; it is not needed here.

5. Worked rows for $W_3(2,5)$

Example 5.1 ($W_3(2,5)$, 414 tiles)**.** $(e,f,M) = (2,5,3)$: $a = 10$, $b = 21$, $c = 25$, $\kappa = 46$, $d = 3$; $W_3(2,5)$ has sides $Mfb = 315$, $Me\kappa = 276$, $Mf^3 = 375$. $n = \lfloor 6/5 \rfloor = 1$, $\rho = 1$, $\delta = 0$, $(t,r,q) = (1,2,1)$. (K): $ft = 5 < eM = 6 < fr = 10$, $er = 4 < fq = 5$, $eq = 2 < ft = 5$. Scales: $v = 11$, $s_1 = 5$, $s_2 = 4$, $x = 3$, $u_1 = 3$, $u_2 = 1$. Points (Table 3), in (ξ, y): $P = (0,0)$, $O = (315,0)$, $B = (0,126)$, $A = (0,-150)$, $P_2 = (42,0)$, $P_3 = (0,-20)$, $Q = (0,105)$, $R = (0,210)$, $C' = (84,0)$, $B' = (40,110)$, $V_2 = (50,85)$, $X_4 = (105,0)$, $U_1 = (30,30)$, $U_2 = (80,10)$; in (u,w): $O = (7.5,3)$, $B = (1.2,3)$, $P_2 = (1,0.4)$, $Q = (1,2.5)$, $C' = (2,0.8)$, $B' = (2,3)$, $V_2 = (2,2.5)$, $U_1 = (1,1)$, $U_2 = (2,1)$.

	scales	formula	tiles		scales	formula	tiles
π_0	15 minus 2	$225-4$	221	π_3	5 minus 4	$25-16$	9
π_1	11	11^2	121	π_4	$as_1 \times cx$	$2\cdot 5\cdot 3$	30
π_2	5	5^2	25	π_5	3 minus 1	$9-1$	8

Total $221+121+25+9+30+8 = 414 = 9\cdot 46 = M^2(2f^2-e^2)$. The tiling is drawn in Figure 4 (numbers in circles are the piece numbers); a list of the 414 tiles in exact coordinates is shipped and verified by an exact checker.

Three rows of the proof, by hand, for this example. *Separation* ($\pi_4|\pi_5$ by the line $w = q$). $w = \psi/(efb) = (2\xi + 5y)/210$. On π_4: $w(Q) = 525/210 = 2.5$, $w(U_1) = (60+150)/210 = 1$, $w(U_2) = (160+50)/210 = 1$, $w(V_2) = (100+425)/210 = 2.5$: all $\geq 1 = q$. On π_5: $w(U_1) = 1$, $w(P_2) = 84/210 = 0.4$, $w(C') = 168/210 = 0.8$, $w(U_2) = 1$: all ≤ 1. So the line $w = 1$ separates the two (the pieces meet along U_1U_2). *Containment* (π_5, vertex U_2, in $\Omega = OAB$). The three sides: $\xi \geq 0$: $80 \geq 0$; $w \leq M$: $w(U_2) = 1 \leq 3$; $ef\xi - by \leq ef^2bM$: $10\cdot 80 - 21\cdot 10 = 590 \leq ef^2bM = 10\cdot 5\cdot 21\cdot 3 = 3150$. Equivalently, as in Step 1: $U_2 \in [C', B']$, since on $u = 2$ its level $w = 1$ lies in $[0.8, 3]$. *Convexity and orientation* ($\pi_5 = U_1P_2C'U_2$). The edge vectors in (ξ, y) are $U_1 \to P_2 = (12,-30)$, $P_2 \to C' = (42,0)$, $C' \to U_2 = (-4,10)$, $U_2 \to U_1 = (-50,20)$, and the consecutive cross products $\xi y' - y\xi'$ are $12\cdot 0 - (-30)\cdot 42 = 1260$, $42\cdot 10 - 0 = 420$, $(-4)(20) - 10(-50) = 420$, $(-50)(-30) - 20\cdot 12 = 1260$: all positive. (In the proof convexity is automatic, since each piece is an intersection of half-planes.) *A shape row:* $P_2 - U_1 = (12,-30) = eu_1(e,-f) = 6\,(2,-5)$, of length $6f = 30 = a\,u_1$ since $au_1 = 10\cdot 3$. The first four columns of the earlier version's tables (Appendix A.3) contain the same facts in the form used by the Lean proof.

Example 5.2 ($W_4(3,5)$, 656 tiles: below the old threshold, $M = 4 < 5 = f$)**.** $(e,f,M) = (3,5,4)$: $a = 15$, $b = 16$, $c = 25$, $\kappa = 41$; $n = \lfloor 12/5 \rfloor = 2$, $\rho = 2$, $\delta = 0$, $(t,r,q) = (2,3,2)$; $v = 11$, $s_1 = 5$, $s_2 = 3$, $x = 4$, $u_1 = 4$, $u_2 = 1$; counts $364+121+100+16+40+15 = 656 = 16\cdot 41$. See Figure 5.

Example 5.3 (the case $\delta = 1$: $(e,f,M) = (2,5,5)$, $M = f$)**.** Here $f \mid eM$: $n = 2$, $\rho = 0$, $\delta = 1$, so $(t,r,q) = (1,3,2)$ and $v = 19$, $s_1 = 10$, $s_2 = 5$, $x = 1$, $u_1 = 8$, $u_2 = 4$; counts $621+361+25+75+20+48 = 1150 = 25\cdot 46$. (K): $5 < 10 < 15$, $6 < 10$, $4 < 5$. Here the strip is two tile-rows wide ($r - t = 2$).

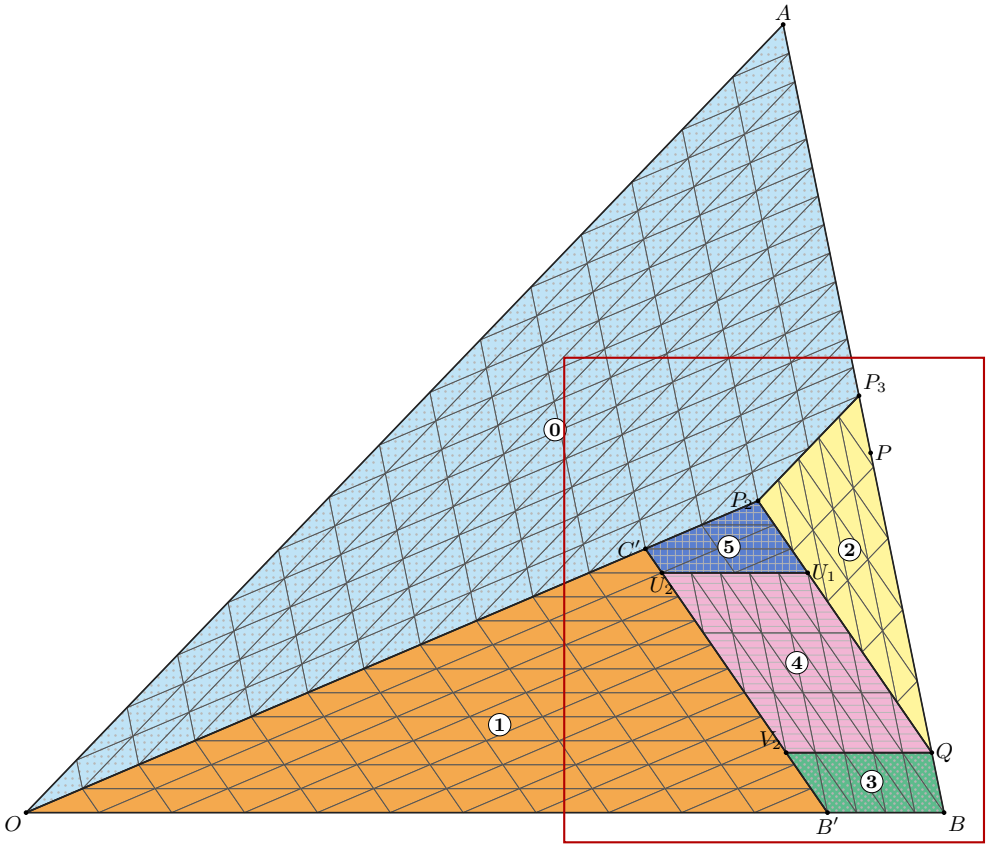
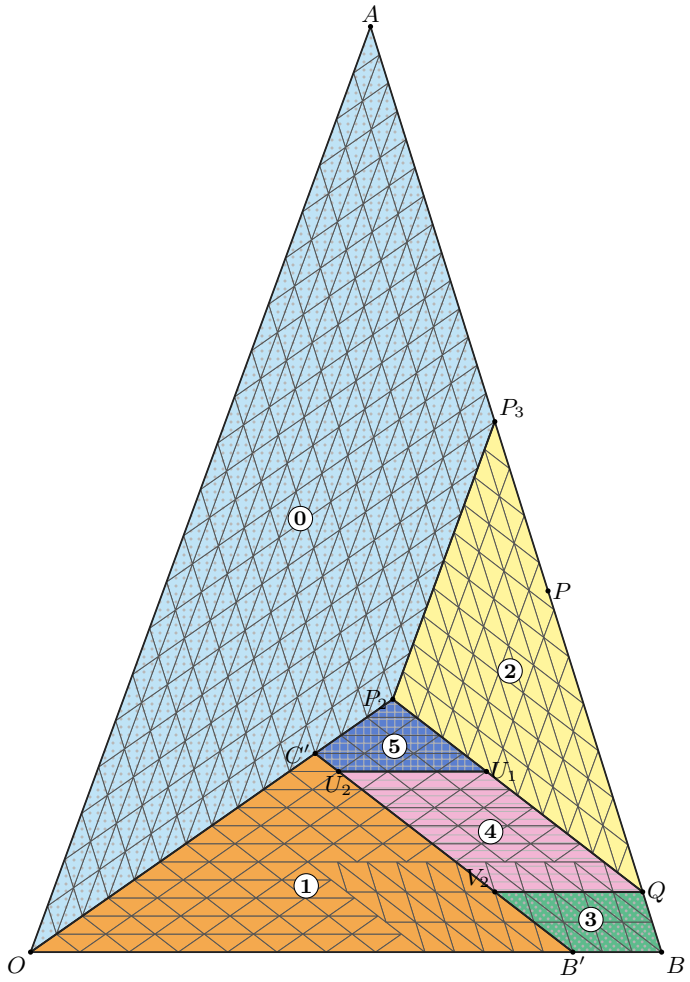


FIGURE 4. The layout for $W_3(2,5)$ (tile $(10,21,25)$, 414 tiles, $(M,n,t,r,q) = (3,1,1,2,1)$). Colours are pieces, numbered 0–5 as in Table 4 (0 light blue, 1 orange, 2 yellow, 3 green, 4 pink, 5 dark blue); the thin lines are the 414 tiles of the exact tiling. The large pieces π_0 and π_1 carry 342 of the 414 tiles; the four small pieces lie in the red box (compare Figure 3).

Example 5.4 (no layout: $(e,f,M) = (3,5,3)$)**.** $d = 2$, $n = \lfloor 9/5 \rfloor = 1$, $dn = 2 \le 3 = e$, so by Proposition 4.1 there are no integers t, r, q with (K): the canonical triple $(1,2,1)$ gives $u_2 = fq - er = 5 - 6 = -1 < 0$. This does not show that $W_3(3,5)$ is untiled (it is an open case, Section 6).

6. Consequences, and what stays open

6.1. **Effect on the census of tile counts.** Let F be the set of those numbers $M^2(2f^2-e^2)$ and $M^2(3f^2-e^2)$, for the triples (e,f,M) of Theorem 3.1, that are non-squarefree and $\le 10^6$. By the theorem all of them are tile counts. The census of the companion covers the non-squarefree integers N with $2 \le N \le 10^6$ (the squarefree ones are treated separately there and are not part of this count). Among them the set F has 44 495 elements (28 946 of the form W, 18 431 of the form T, with 2 882 values of both forms; largest f is 251), none of them is excluded by the exclusion theorem of the earlier release, 18 357 were known tile counts by other constructions, and 26 138 **moved from "undecided" (neither known to be a tile count nor excluded by the exclusion theorem of the earlier release) to "tile count"** (each such value is an instance of the Lean theorem `thmW`, which is universal in (e,f,M); the count 26 138 itself comes from a program, [COMPUTED, EXACT], and the "undecided" status comes from the companion



0 $\triangle OAP$ of scale fM minus corner of scale et: $f^2M^2 - e^2t^2 = 364$

1 triangle $OB'C'$ of scale $v = fM - er$: $v^2 = 121$

2 triangle of scale ft: $f^2t^2 = 100$

3 triangle of scale $s_1 = f(r-t)$ minus corner of scale $s_2 = fr - eM$: $s_1^2 - s_2^2 = 16$

4 parallelogram $as_1 \times cx$, $x = ft - eq$: $2s_1x = 40$

5 triangle of scale $u_1 = fq - et$ minus corner of scale $u_2 = fq - er$: $u_1^2 - u_2^2 = 15$

$t = n - \delta$, $\delta = [f \mid eM]$, $r = n + 1$, $q = n$, $n = \lfloor eM/f \rfloor$ here $(M, n, t, r, q) = (4, 2, 2, 3, 2)$ total $= M^2(2f^2 - e^2) = 656$

FIGURE 5. The same layout for $W_4(3, 5)$ (656 tiles of $(15, 16, 25)$), $(M, n, t, r, q) = (4, 2, 2, 3, 2)$). Colours and numbers as in Figure 4; counts $364 + 121 + 100 + 16 + 40 + 15$.

census). The census itself, the other constructions and the conjecture that organises them are in the companion paper [6]. Two examples: $414 = 2 \cdot 3^2 \cdot 23$ and $639 = 3^2 \cdot 71$ were undecided before ($414 = W_3(2, 5)$, $639 = T_3(2, 5)$).

6.2. **What stays open.**

- $e < f/2$**: nothing** (Corollary 3.2) for W_M; for T_M the scales with $2Me \geq f$ and $M < M^*(e, f)$ are neither proved tiled nor excluded (for example $T_3(2, 11)$).

- $e > f/2$, **the "no" side.** The exclusion imported here is only (I2), $M(f-e) \le e$. So for each row (e, f) with $e > f/2$ the scales M neither covered by Theorem 3.1 nor excluded by (I2) are exactly those with $e/d < M < f/d + 1$, $d = f - e$ (for $e > f/2$ the condition $Me \ge f$ holds for all $M \ge 2$); the interval has length exactly 2, so it contains at most two integers: at most two undecided scales per row. (That the interval holds at most two integers is proved: it is an open interval of length 2.) [CONJECTURE] *Conjecture W^*:* for coprime e, f with $2 \le e \le f - 2$, $W_M(e, f)$ is tiled iff $M \ge M^*(e, f)$ (for $e > f/2$: iff $(M-1)(f-e) \ge f$). Examples, relative to the imports (I1)–(I3) only (see also the next item): $W_2(3, 5)$, $W_3(3, 5)$, $W_2(5, 8)$. For $W_2(5, 8)$ ($412 = 4 \cdot 103$ tiles) a restricted exact computer search found no tiling (evidence only, not formal, [COMPUTED, EXACT]); the next item says it is decided by a newer theorem.
- **A newer exclusion, not used in this paper.** In the project repository (not in the release that accompanies this paper) there is a Lean theorem, `Draft.SplusThick.wTilable_not_of_small_M`, stating that $W_M(e, f)$ is not tiled when $\gcd(e, f) = 1$, $2 \le e < f$, $M < f$, $f \le Me$ and $M(f-e) \le e+1$; it extends (I2) by one and decides $W_2(3, 5)$ and $W_2(5, 8)$ (no tiling) and leaves $W_3(3, 5)$ open. Its hand proof has been audited inside the project (verdict: sound, with one repair), the Lean file is marked unaudited in its header, and it has not been checked by anyone outside the project. It does not interact with Theorem 3.1: for $e > f/2$ Theorem 3.1 needs $(M-1)(f-e) \ge f$, the exclusion needs $M(f-e) \le e + 1 < f$.
- $e = 1$ **and** $e = f - 1$ lie outside Theorem 3.1; for W_M the threshold is $M = f$ by (I3); for $T_M(1, f)$ the exclusion $2M \le f$ of (I1) leaves, for example, $T_3(1, 4)$ open.
- **The T half in general:** only the tile count and the glue; the converse of the glue is unknown.
- **Nothing is claimed about the prime case or any integer outside F.**

Outside check. The Lean statements and proofs were compared and audited inside this project only; no reader outside the project has checked them.

7. Credits and disclosure

Credit. The targets W_M, T_M, the W/β framework and the tilings for $M \ge f$ are due to D. Paliy and to M. Beeson and G. Roşu (the tile with $3\alpha + 2\beta = \pi$ is Beeson's case III [1]); the isosceles-apex frame at the point P with $OP = OB$ is the one of Beeson–Roşu; the six-piece layout was found by our own computer search and is proved here, and the lattice lines ℓ_k of Section 2.4 are our way of reading it. The companion paper [6] credits, statement by statement, B. Herdt, Y. X. Zhang, M. Laczkovich, J. P. Harries, and Snover–Waiveris–Williams, whose work enters there rather than here. *AI.* The work was done with the help of an AI system (Claude, Anthropic); the questions and directions were the author's, and each claim carries the label saying how it was checked (explicit tilings are checked by an exact program, not by Lean). The statements "Paliy" in this paper refer to his public repository [3]; we have read his documents and checked some examples; nothing here claims priority over him for the results listed as his.

Appendix A. Verification aid: what was computed, and the older proof

This appendix is not needed to follow Section 4.

A.1. What is checked how.

statement	by what
Frame: coordinates of Table 3, the metric (1), the five directions	hand (Section 2.4); exact arithmetic against the older complex-number description (Appendix A.3) for all coprime e, f with $f \le 24$, $M < 3f$ (all $f \le 40$ with sampled M), canonical and random (t, r, q): no mismatch (`code/check_thmW_frame.py`, block A)
Side lengths of Table 5, parallelism, levels of the cut lines	hand; exact: 18 276 cases, no mismatch (block B)
Step 1 (six pieces: separation, containment, convexity, orientation, areas add up)	hand; exact: 201 886 triples (e, f, M), $f \le 60$, $M < 5f$: no failure under (H1), (H2); none passes outside (block D)
(K) $\Leftrightarrow$ the 16 sign conditions of the older proof; layout exists iff $dn \ge e+1$; canonical triple works whenever any works	hand (Prop. 4.1); exact enumeration of all (t, r, q) for $f \le 12$, $M < 3f$, 17.9 million triples (block E)
Lemma 3.5 and every step of its proof	hand; exact: all coprime $2 \le e \le f-2$, $f \le 300$, $M \le 3f$, 16.2 million triples (block G)
Count identity of Section 4.5, and the three factor identities $C_8 = fC_7 + eC_6$, $C_9 = fC_2 + C_7$, $C_6' = C_6 + eC_2$	computer algebra (block H, sympy)
The Lean proof (the older route)	Lean (`ThmWIneq0`–`4`, `ThmWGeo0`–`3`, `ThmWParams`, `ThmWNat`, `ThmWFinal`); the cross-product tables of Appendix A.3 are produced by `code/check_thmW_algebra.py`
Packing: finitely many convex pieces, each tiled, disjoint interiors, area sum = area of target $\Rightarrow$ tiling	Lean (general theorem in the library)
Explicit tilings (e.g. 414, 656, 639, 801, 3950, 13 888, 57 232 tiles)	exact checker on exact coordinates: congruence, disjointness, area, hull a triangle ([COMPUTED, EXACT]; not Lean)
Not in this paper's Lean statement	the tiling of T_M (glue (I5), Lean in the earlier release); the exclusions (I1)–(I3)

Consistency with the proved exclusions: (H1), (H2) imply $M(f-e) > e$, $Me \ge f$ and $2Me \ge f$, so no instance of the theorem contradicts (I1) or (I2). Reproduction: `python3 code/check_thmW_frame.py ALL 60` (about one minute on one core) and `python3 code/check_thmW_algebra.py 60 -tex tables.tex`.

A.2. **Alternative to Step 1: the chain lemma, and certificates for (K).** *Hand-proved, audited inside the project (verdict: sound, with four repairs, all applied here); not in Lean.* Step 1 above is a direct case analysis. A shorter argument replaces containment, separation and the area identity by one lemma (winding-number additivity; the integer frame is due to a study inside this project).

Lemma A.1 (chain lemma)**.** *Let T and $\Pi_1, \dots, \Pi_k$ be polygons with counter-clockwise vertex lists, all with* strictly *positive turns (cross products of consecutive edge vectors). Then each Π_i is a convex polygon: for a* 3- *or* 4*-gon whose exterior angles all lie in $(0, \pi)$ and sum to 2π, the boundary is a simple closed convex curve. If, as formal sums of oriented segments, $\sum_i \partial\Pi_i = \partial T$, where the equality means equal oriented multiplicity on every line (collinear segments add:*

$[p,q]+[q,r]=[p,r]$ *in any order of the points, and* $[p,q]+[q,p]=0$*), then the* Π_i *have pairwise disjoint interiors, lie in* T*, and cover* T*.*

Proof. Winding numbers are additive in the boundary chain; a convex counter-clockwise polygon has winding number 1 inside and 0 outside, so $\sum_i \mathbf{1}_{\Pi_i} = \mathbf{1}_T$ off finitely many lines. If two interiors met, the sum would be ≥ 2 on an open set; if some Π_i left T, it would be ≥ 1 off T. □ (Caution: "each piece inside T and the areas add up" does *not* imply disjointness; and the vanishing of a sum of edge *vectors* is not chain equality.) *The chain identity* holds for all integers e, f, M, t, r, q, because the points of Table 3 lie on five lines: AB ($\xi = 0$: A, P_3, Q, B), OP ($y = 0$: O, P_2, C'), OB ($e\xi + fy = efbM$: O, B, B'), ℓ_t ($\varphi = efbt$: P_2, Q, U_1), ℓ_r ($\varphi = efbr$: C', B', U_2, V_2); in each the parameters cancel (for U_1: $f \cdot ef(ft - eq) + e \cdot ef(fq - et) = efbt$). Then on each line the oriented edges cancel: $C' \to O + O \to P_2 + P_2 \to C' = 0$; $B \to Q + Q \to P_3 + P_3 \to A = B \to A$; $O \to B' + B' \to B = O \to B$; $P_2 \to Q + Q \to U_1 + U_1 \to P_2 = 0$; $V_2 \to B' + B' \to C' + C' \to U_2 + U_2 \to V_2 = 0$; and P_2P_3, QV_2, U_1U_2 are shared by two pieces with opposite orientation. *What remains* is strict positivity of every turn: each is a positive monomial times a product of the factors t, $r - t$, $eM - ft$, $fr - eM$, $fM - er$, $ft - eq$, $fq - er$, $fM - et$, $fq - et$. Five are independent: t, $C_3 = eM - ft$, $C_4 = fr - eM$, $C_6 = fq - er$, $C_7 = ft - eq$; the others follow from $f(r - t) = C_3 + C_4$, $fq - et = C_6 + e(r - t)$, $fM - et > 0$ (as $et < eM < fM$) and the identity $e\,C_5 = f\,C_3 + f\,C_7 + e\,C_6$ for $C_5 = fM - er$ (and $C_5 = u + 1 + fs$ below, positive anyway). *Certificates.* Put $\mu := dn - e - 1$ (≥ 0 by Lemma 3.5), $\rho' = f - 1 - \rho \geq 0$, $M = n + s$, so $\rho = es - dn$. For the canonical triple: $C_3 = \rho + f\delta$, $C_4 = \rho' + 1$, $C_5 = \mu + 1 + fs$, $C_6 = \mu + 1$, and $C_7 = \mu + 1 + e$ if $\delta = 0$; if $\delta = 1$ ($M = fm'$, $n = em'$, $e - 1 = 1 + x$, $d - 1 = 1 + y$, $x, y \geq 0$, $x + y \geq 1$) then $C_7 = (m' - 1)ed + x + y + xy \geq 1$. Each is a sum of non-negative terms with a positive constant. Checked exactly for all 982 658 admissible triples ($f \leq 120$, $M \leq 3f$); the 27 322 non-admissible triples in that range each violate a turn, so for these triples the canonical layout fails (compare Proposition 4.1, which is the actual sharpness statement). The Lean proof of this paper does not use the chain lemma: it proves the placement conditions the older way (the sixteen named factors of Appendix A.3, 143 generated sign lemmas, kernel-checked). A Lean version of the chain lemma is planned, not done. Still needed for Step 2, unchanged: that each piece is a union of tiles (the edge-length identities of Table 5 and Proposition 2.1).

A.3. **The older proof: complex coordinates and cross-product tables.** The first version of this paper, and the Lean proof, use complex numbers with $O = 0$ and OB the positive real axis. With $D = 4f^2 - e^2$ write $\xi + i\sqrt{D}\,\zeta$ as $[\xi, \zeta]$ and put

$$\varepsilon = e^{i\alpha} = \left[\tfrac{\kappa}{2f^2}, \tfrac{e}{2f^2}\right], \quad \eta = e^{i\beta} = \left[\tfrac{e(3f^2 - e^2)}{2f^3}, \tfrac{b}{2f^3}\right], \quad \omega = \overline{\varepsilon\eta} = e^{-i(\alpha+\beta)},$$

with $\varepsilon^3\eta^2 = -1$, $a + b\omega = c\bar{\eta}$, $f\omega + e\varepsilon = f\bar{\eta}$ (each checked by expansion). Then $P = fbM\varepsilon$ and the frame is $z = P - \xi\,\varepsilon + y\,\omega$: the first axis points from P to O (direction $-\varepsilon$) and the second from P to B (direction ω). In these terms the points of Table 3 are $B = fbM$, $A = f^3M\varepsilon^2$, $P_2 = b(fM - et)\varepsilon$, $P_3 = P - eat\,\omega$, $Q = P + fbt\,\omega$, $R = P + fbr\,\omega$, $C' = bv\,\varepsilon$, $B' = cv$, $V_2 = P + f^3t\,\bar{\eta} - ef^2r$, $X_4 = P - fbq\,\varepsilon$, $U_1 = P - ef^2t + ef^2q\,\bar{\eta}$, $U_2 = P - ef^2r + ef^2q\,\bar{\eta}$ (exact check against the frame: block A). The cross product $\langle u, v\rangle = \frac{1}{\sqrt{D}}\mathrm{Im}(\bar{u}v)$ is $1/\sqrt{D}$ times the usual one. The Lean proof shows that each cross product needed for the containment of a vertex in OAB, for the separation of two pieces and for the convexity of a piece equals a positive

constant times a product of the *named factors*

$$C_2 = r - t,\ C_3 = eM - ft,\ C_3' = fM - et,\ C_4 = fr - eM,\ C_5 = fM - er,$$
$$C_6 = fq - er,\ C_6' = fq - et,\ C_7 = ft - eq,\ C_8 = f^2t - e^2r,\ C_9 = fr - eq,$$
$$C_{10} = M - q,\ E_1 = bM + eft,\ E_2 = eMf + bt,\ E_3 = br + e^2t,$$
$$L_A = bfM - e\kappa t + f^3q,\ L_B = bfM - e\kappa r + f^3q,\ L_C = efbM - e^2\kappa r + f^4t,$$

and that these are positive for the canonical triple. Under (K) they all follow from the four inequalities: $C_3, C_4, C_6, C_7 > 0$ are (K) itself; $C_2 > 0$, $C_{10} > 0$ and $C_3', C_5 > 0$ are (4); $C_6' = C_6 + eC_2$, $C_8 = fC_7 + eC_6$, $C_9 = fC_2 + C_7$ are sums of positive terms; E_1, E_2, E_3 are sums of products of positive numbers; L_A, L_B, L_C express that U_1, U_2, V_2 lie on the segments $[P_2, Q]$, $[C', B']$ of Step 1, which is automatic (verified exactly: block E). The tables, generated by `code/check_thmW_algebra.py`, are: The values of the sixteen factors of the older proof for the three examples (all positive):

(e,f,M)	C_2	C_3	C_3'	C_4	C_5	C_6	C_6'	C_7	C_8	C_9	C_{10}	L_A	L_B	L_C
$(2,5,3)$	1	1	13	4	11	1	3	3	17	8	2	348	256	887
$(3,5,4)$	1	2	14	3	11	1	4	4	23	9	2	324	201	1103
$(2,5,5)$	2	5	23	5	19	4	8	1	13	11	3	683	499	1123

Appendix B. Data, Lean index, reproduction

B.1. **The thirteen witnesses with $M < f$ found by the first search.** Found first by a computer search, now instances of Theorem 3.1; each is a Lean theorem (module `Draft/DebtBWTilable`) with the closure upward in M (I5). Columns: tiles of W_M and of T_M.

(e,f,M)	W_M	T_M	(e,f,M)	W_M	T_M
$(3,5,4)$	656	1056	$(7,12,4)$	3824	6128
$(4,7,4)$	1312	2096	$(7,10,5)$	3775	6275
$(5,8,4)$	1648	2672	$(7,9,6)$	4068	6984
$(5,7,5)$	1825	3050	$(8,11,5)$	4450	7475
$(5,11,3)$	1953	3042	$(9,11,7)$	7889	13 818
$(5,9,4)$	2192	3488	$(6,11,4)$	3296	5232
$(7,11,4)$	3088	5024			

B.2. **Shipped tilings.** Directory `data/witnesses/v3_exact/`; files are named by tile count, kind and parameters (for example `tiling_414_W_2_5_M3`, `tiling_656_W_3_5_M4`, `tiling_639_T_2_5_M3`, `tiling_1056_T_3_5_M4`, each with the suffix `.exact.txt.gz`). Each begins with a comment line (tile count, tile sides); each further line gives the three vertices of one tile in exact coordinates $(X, Y\sqrt{D})$. Kinds `W` and `T`: 37 and 15 files.

B.3. **Lean index.** All names in namespace `Erdos634.`; aggregate `AllThmW`. Theorem 3.1: `Draft.ThmW.thmW` (modules `ThmWLib`, `Lib2`, `Hyp`, `Sc`, `Pts`, `Ineq0`–`4`, `Geo0`–`3`, `Core`, `Nat`, `Params`, `Final`); instances `inst_414`, `inst_656`, `inst_639`, `inst_801`, `inst_3088`, `inst_27324`. Counts: `Draft.DebtB.counts_of_wTilable`; glue: `CollarG.betaTilable_of_wTilable`; 13 witnesses: `Draft.DebtB.closure_656` etc.; ladder: `Subdivision.ladderDissection`.

B.4. **Reproduction.**

```
python3 code/check_thmW_frame.py ALL 60                # frame, layout, equivalences
python3 code/check_thmW_algebra.py 60 --tex tables.tex  # older algebra + tables
python3 code/lakeserial.py lean build Erdos634.AllThmW  # one module at a time
BTE=rust/bonfioli_tiling_engine/target/release/bte
```

TABLE 6. Separation of pieces: cross products, one row for each vertex of the second piece not on the edge line.

pair	edge line	vertex	$-$cross = positive constant $\times$
$\pi_0 \vert \pi_1$	OP_2	B'	$\frac{1}{2}\, C_3'\, C_5\, d\, e\, (f{+}e)$
$\pi_0 \vert \pi_2$	P_2P_3	Q	$\frac{1}{2}\, d\, e\, f^2\, (f{+}e)\, t^2$
$\pi_0 \vert \pi_3$	OP_2	B	$\frac{1}{2}\, C_3'\, M\, d^2\, e\, f^{-1}\, (f{+}e)^2$
$\pi_0 \vert \pi_3$	OP_2	Q	$\frac{1}{2}\, C_3'\, d^2\, (f{+}e)^2\, t$
$\pi_0 \vert \pi_3$	OP_2	V_2	$\frac{1}{2}\, C_3'\, C_8\, d\, (f{+}e)$
$\pi_0 \vert \pi_3$	OP_2	B'	$\frac{1}{2}\, C_3'\, C_5\, d\, e\, (f{+}e)$
$\pi_0 \vert \pi_4$	OP_2	Q	$\frac{1}{2}\, C_3'\, d^2\, (f{+}e)^2\, t$
$\pi_0 \vert \pi_4$	OP_2	U_1	$\frac{1}{2}\, C_3'\, C_6'\, d\, e\, (f{+}e)$
$\pi_0 \vert \pi_4$	OP_2	U_2	$\frac{1}{2}\, C_3'\, C_6\, d\, e\, (f{+}e)$
$\pi_0 \vert \pi_4$	OP_2	V_2	$\frac{1}{2}\, C_3'\, C_8\, d\, (f{+}e)$
$\pi_0 \vert \pi_5$	OP_2	U_1	$\frac{1}{2}\, C_3'\, C_6'\, d\, e\, (f{+}e)$
$\pi_0 \vert \pi_5$	OP_2	U_2	$\frac{1}{2}\, C_3'\, C_6\, d\, e\, (f{+}e)$
$\pi_1 \vert \pi_2$	$B'C'$	P_3	$\frac{1}{2}\, C_5\, E_3\, e^2$
$\pi_1 \vert \pi_2$	$B'C'$	P_2	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_1 \vert \pi_2$	$B'C'$	Q	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_1 \vert \pi_3$	$B'C'$	B	$\frac{1}{2}\, C_4\, C_5\, d\, e^2\, f^{-1}\, (f{+}e)$
$\pi_1 \vert \pi_3$	$B'C'$	Q	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_1 \vert \pi_4$	$B'C'$	Q	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_1 \vert \pi_4$	$B'C'$	U_1	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_1 \vert \pi_5$	$B'C'$	U_1	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_1 \vert \pi_5$	$B'C'$	P_2	$\frac{1}{2}\, C_2\, C_5\, d\, e^2\, (f{+}e)$
$\pi_2 \vert \pi_3$	P_2Q	B	$\frac{1}{2}\, C_3\, d^2\, e\, f^{-1}\, (f{+}e)^2\, t$
$\pi_2 \vert \pi_3$	P_2Q	V_2	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_2 \vert \pi_3$	P_2Q	B'	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_2 \vert \pi_4$	P_2Q	U_2	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_2 \vert \pi_4$	P_2Q	V_2	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_2 \vert \pi_5$	P_2Q	C'	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_2 \vert \pi_5$	P_2Q	U_2	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_3 \vert \pi_4$	QV_2	U_1	$\frac{1}{2}\, C_2\, C_7\, d\, e\, f\, (f{+}e)$
$\pi_3 \vert \pi_4$	QV_2	U_2	$\frac{1}{2}\, C_2\, C_7\, d\, e\, f\, (f{+}e)$
$\pi_3 \vert \pi_5$	QV_2	U_1	$\frac{1}{2}\, C_2\, C_7\, d\, e\, f\, (f{+}e)$
$\pi_3 \vert \pi_5$	QV_2	P_2	$\frac{1}{2}\, C_2\, d^2\, e\, (f{+}e)^2\, t$
$\pi_3 \vert \pi_5$	QV_2	C'	$\frac{1}{2}\, C_2\, C_8\, d\, e\, (f{+}e)$
$\pi_3 \vert \pi_5$	QV_2	U_2	$\frac{1}{2}\, C_2\, C_7\, d\, e\, f\, (f{+}e)$
$\pi_4 \vert \pi_5$	U_1U_2	P_2	$\frac{1}{2}\, C_2\, C_6'\, d\, e^2\, (f{+}e)$
$\pi_4 \vert \pi_5$	U_1U_2	C'	$\frac{1}{2}\, C_2\, C_6\, d\, e^2\, (f{+}e)$

```
$BTE verify data/witnesses/v3_exact/tiling_414_W_2_5_M3.exact.txt.gz
```

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[2] M. Beeson and G. Roşu, *Tiling a triangle into a prime number of congruent triangles*, preprint, version of 16 October 2026.

[3] D. Paliy, repository `github.com/DenisUkranian/erdos-634-tilings`: the W/β framework, caps and collars, tilings for $M \geq f$ (September–October 2026); unrefereed.

TABLE 7. Containment in OAB: $\langle Y-X, Z-X\rangle \geq 0$ for each vertex Z of a piece and each side XY of the target in the order OB, BA, AO (zero rows omitted).

vertex	side line of OAB	cross product = positive constant $\times$
A	OB	$\frac{1}{2}\,M^2\,\kappa\,d\,e\,(f{+}e)$
O	BA	$\frac{1}{2}\,M^2\,\kappa\,d\,e\,(f{+}e)$
P_2	OB	$\frac{1}{2}\,C_3'\,M\,d^2\,e\,f^{-1}\,(f{+}e)^2$
P_2	BA	$\frac{1}{2}\,M\,\kappa\,d\,e^2\,f^{-1}\,(f{+}e)\,t$
P_2	AO	$\frac{1}{2}\,C_3'\,M\,d\,e\,f\,(f{+}e)$
P_3	OB	$\frac{1}{2}\,E_1\,M\,d\,e\,(f{+}e)$
P_3	AO	$\frac{1}{2}\,C_3'\,M\,d\,e\,f\,(f{+}e)$
B'	BA	$\frac{1}{2}\,C_4\,M\,\kappa\,e^2$
B'	AO	$\frac{1}{2}\,C_5\,M\,\kappa\,e\,f$
C'	OB	$\frac{1}{2}\,C_5\,M\,d^2\,e\,f^{-1}\,(f{+}e)^2$
C'	BA	$\frac{1}{2}\,M\,\kappa\,d\,e^2\,f^{-1}\,(f{+}e)\,r$
C'	AO	$\frac{1}{2}\,C_5\,M\,d\,e\,f\,(f{+}e)$
Q	OB	$\frac{1}{2}\,C_3\,M\,d^2\,(f{+}e)^2$
Q	AO	$\frac{1}{2}\,E_2\,M\,d\,f\,(f{+}e)$
B	AO	$\frac{1}{2}\,M^2\,\kappa\,d\,e\,(f{+}e)$
V_2	OB	$\frac{1}{2}\,C_3\,M\,d^2\,(f{+}e)^2$
V_2	BA	$\frac{1}{2}\,C_2\,M\,\kappa\,e^2\,f$
V_2	AO	$\frac{1}{2}\,\mathrm{L}_C\,M\,f$
U_1	OB	$\frac{1}{2}\,C_{10}\,M\,d^2\,e\,(f{+}e)^2$
U_1	BA	$\frac{1}{2}\,C_7\,M\,\kappa\,e^2$
U_1	AO	$\frac{1}{2}\,\mathrm{L}_A\,M\,e\,f$
U_2	OB	$\frac{1}{2}\,C_{10}\,M\,d^2\,e\,(f{+}e)^2$
U_2	BA	$\frac{1}{2}\,C_9\,M\,\kappa\,e^2$
U_2	AO	$\frac{1}{2}\,\mathrm{L}_B\,M\,e\,f$

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[5] V. Bonfioli, *Congruent dissections of a triangle: runs, windows and the corner*, version 2.2, release v1.16, Zenodo (same concept DOI).

[6] V. Bonfioli, *Conjecture C, the collar family and the census of tile counts, with computer evidence* (companion to Paper W), version 3.0, release v3.0.

TABLE 8. Convexity: the consecutive cross products ("turns") of each piece, with the vertex X_j at which each starts; all are positive. The doubled signed area of piece π_i equals $\frac{eb}{2}$ times its number of tiles.

piece	vertex X_j	turn $\langle X_{j+1}-X_j, X_{j+2}-X_{j+1}\rangle$ = positive constant $\times$
π_0	A	$\frac{1}{2}\,C_3'\,M\,d\,e\,f\,(f{+}e)$
π_0	O	$\frac{1}{2}\,C_3'\,d\,e^2\,(f{+}e)\,t$
π_0	P_2	$\frac{1}{2}\,C_3'\,d\,e^2\,(f{+}e)\,t$
π_0	P_3	$\frac{1}{2}\,C_3'\,M\,d\,e\,f\,(f{+}e)$
π_1	O	$\frac{1}{2}\,C_5^2\,d\,e\,(f{+}e)$
π_1	B'	$\frac{1}{2}\,C_5^2\,d\,e\,(f{+}e)$
π_1	C'	$\frac{1}{2}\,C_5^2\,d\,e\,(f{+}e)$
π_2	P_3	$\frac{1}{2}\,d\,e\,f^2\,(f{+}e)\,t^2$
π_2	P_2	$\frac{1}{2}\,d\,e\,f^2\,(f{+}e)\,t^2$
π_2	Q	$\frac{1}{2}\,d\,e\,f^2\,(f{+}e)\,t^2$
π_3	B	$\frac{1}{2}\,C_2\,C_3\,d\,e\,f\,(f{+}e)$
π_3	Q	$\frac{1}{2}\,C_2\,C_3\,d\,e\,f\,(f{+}e)$
π_3	V_2	$\frac{1}{2}\,C_3\,C_4\,d\,e\,(f{+}e)$
π_3	B'	$\frac{1}{2}\,C_3\,C_4\,d\,e\,(f{+}e)$
π_4	Q	$\frac{1}{2}\,C_2\,C_7\,d\,e\,f\,(f{+}e)$
π_4	U_1	$\frac{1}{2}\,C_2\,C_7\,d\,e\,f\,(f{+}e)$
π_4	U_2	$\frac{1}{2}\,C_2\,C_7\,d\,e\,f\,(f{+}e)$
π_4	V_2	$\frac{1}{2}\,C_2\,C_7\,d\,e\,f\,(f{+}e)$
π_5	U_1	$\frac{1}{2}\,C_2\,C_6'\,d\,e^2\,(f{+}e)$
π_5	P_2	$\frac{1}{2}\,C_2\,C_6\,d\,e^2\,(f{+}e)$
π_5	C'	$\frac{1}{2}\,C_2\,C_6\,d\,e^2\,(f{+}e)$
π_5	U_2	$\frac{1}{2}\,C_2\,C_6'\,d\,e^2\,(f{+}e)$